

Autodesk® Scaleform®

Scaleform 3D

Scaleform 3.2

3D

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Autodesk® Scaleform® 4.4

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1

Scaleform® 3Di™

UI

3D Flash

Scaleform 3.2

Scaleform

ActionScript

3D

Scaleform 3.2

ActionScript 2.0

3D AS

AS 3.0

HUD

_z
UI

_zscale

_xrotation
3D

_yrotation
3D

_matrix3d

_perspfov

C++

3D

Scaleform Direct Access API

API

3D Gfx::Value

3D

3D Flash

2 Flash 10

Scaleform
Scaleform

3Di
AS2

Flash 10

3D

3Di

2.1

Flash 10

3D

3Di

1.

2. (Back-face culling)

3 3D Flash

Scaleform 3.2

3D

Flash

3.1

Flash

3D

Scaleform 3.2

3D

Flash

“

” (Render to Texture)

3Di

Flash

3D

3D

3D

Z

/

UI

Flash

Scaleform 3.2 (o)6(()Tf0 Tc 0 Tw 4.19 0 T45C102E07A70E2211<173F02-

4 3Di API

4.1 ActionScript 2 API

3Di Flash

z 3d 3d (Null) 3D x y 3D
null 2D ActionScript foo._matrix3d =
z 0 3D
3D 3 x y z Z 2D
3D _zscale ActionScript
Scaleform 3D Flash 10 +X +Y
+Z +Z FOV 55

- 3D Flash
-
-
-

3Di ActionScript

- **_z** - Z 0
- **_zscale** - Z 100
- **_xrotation** - X 0
- **_yrotation** - Y 0
- **_matrix3d** - 16 4 x 4 3D
NULL 3D 2D
- **perspfov** — (FOV) > 0 < 180
-1 FOV
55

ActionScript gfxExtensions True

_xrotation _yrotation 3D

1

Y 45

```
_global.gfxExtensions = true;
mc1._yrotation = 45;
```

2

X Z

```
_global.gfxExtensions = true;
sq1._zscale = 500;

function rotateAboutXAxis(mc:MovieClip):Void
{
    mc._xrotation = mc._xrotation + 1;
    if (mc._xrotation > 360)
    {
        mc._xrotation = 0;
    }
}

onEnterFrame = function()
{
    rotateAboutXAxis(sq1);
}
```

3

3D 3D

```
function PixelsToTwips(iPixels):Number
{
    return iPixels*20;
}
```

```
function TranslationMatrix(tX, tY, tZ):Array
{
    var matrixC:Array = [ 1, 0, 0, 0,
                          0, 1, 0, 0,
                          0, 0, 1, 0,
                          tX,tY,tZ,1];
    return matrixC;
}

this._matrix3d = TranslationMatrix(PixelsToTwips(100),PixelsToTwips(100),0);
```

4.2 C++ 3Di

4.2.1 Matrix4x4

Matrix4x4

4x4

3D

Render_Matrix4x4.h

4.2.2 Direct Access

Direct Access(3ct)-5(A)--0.004F0<<1908>11epF-ZÈÖ-Ç,À8VÂ@ac02EB6L5.2.2B:D Oppit AcMx4

```
void    SetProjectionMatrix3D(const Matrix4F& m)
void    SetViewMatrix3D(const Matrix3F& m);
```

4.2.4 Direct Access API

```
Ptr<GFx::Movie> pMovie = ...;
GFx::Value tmpVal;
bool bOK = pMovie->GetVariable(&tmpVal, "_root.Window");
if (bOK)
{
    GFx::Value::DisplayInfo dinfo;
    bOK = tmpVal.GetDisplayInfo(&dinfo);
    if (bOK)
    {
        // set perspectiveFOV to 150 degrees
        Double oldPersp = dinfo.GetFOV();
        dinfo.SetFOV(150);
        tmpVal.SetDisplayInfo(dinfo);
    }
}
```

5 透视

Direct Access API

(FOV)

FOV 0

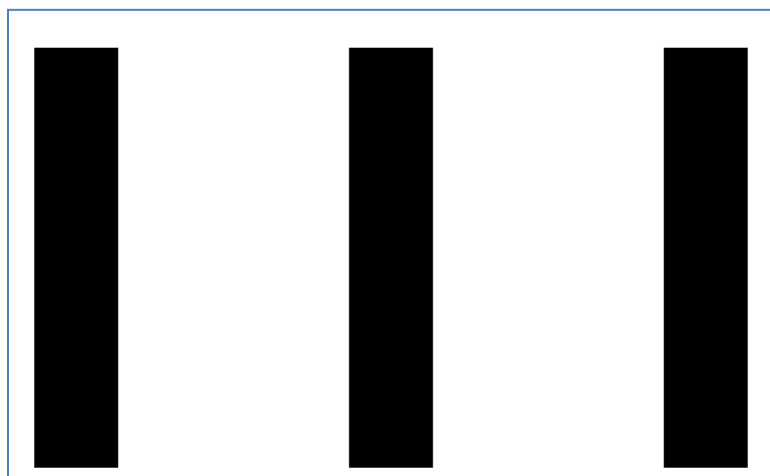
FOV

z

FOV

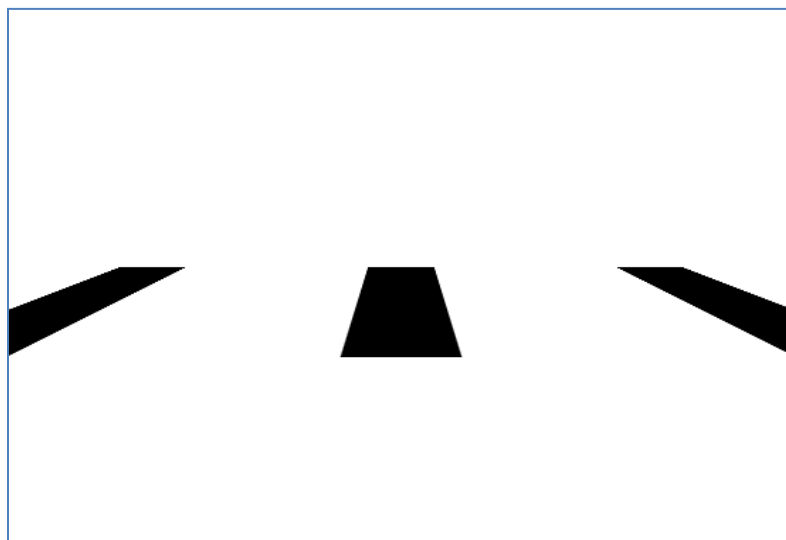
0 180 1

179



2

Flash



3 -80

55

X

FOV (5)
 ActionScript property _perspfov -1 C++
 Direct Access API :

// 1. Perspective FOV -1

```
Gfx::Value tmpVal;
bool bOK = pMovie->GetVariable(&tmpVal, "root"); //"_root" for AS2
if (bOK)
{
  Gfx::Value::DisplayInfo dinfo;
  bOK = tmpVal.GetDisplayInfo(&dinfo);
  if (bOK)
  {
    dinfo.SetPerspFOV(-1);
    tmpVal.SetDisplayInfo(dinfo);
  }
}
```

// 2.

```
void MakeOrthogProj(const RectF &visFrameRectInTwips, Matrix4F *matPersp)
{
  const float nearZ = 1;
  const float farZ = 100000;
  float DisplayHeight = fabsf(visFrameRectInTwips.Height());
  float DisplayWidth = fabsf(visFrameRectInTwips.Width());
  matPersp->OrthoRH(DisplayWidth, DisplayHeight, nearZ, farZ);
}
Gfx::Value tmpVal;
bool bOK = pMovie->GetVariable(&tmpVal, "root.mc");
if (bOK)
{
  Gfx::Value::DisplayInfo dinfo;
  bOK = tmpVal.GetDisplayInfo(&dinfo);
  if (bOK)
  {
    Matrix4F perspMat;
    MakeOrthogProj(PixelsToTwips (pMovie->GetVisibleFrameRect()), &perspMat);
    dinfo.SetProjectionMatrix3D(&perspMat);
    tmpVal.SetDisplayInfo(dinfo);
  }
}
```

(Projection)

```

Ptr<Movie> pMovie = ...
// compute ortho matrix
Render::Matrix4F ortho;
const RectF &visFrameRectInTwips = PixelsToTwips(pMovie->GetVisibleFrameRect());
const float nearZ = 1;
const float farZ = 100000;
ortho.OrthoRH(fabs(visFrameRectInTwips.Width()),  fabs(visFrameRectInTwips.Height()),
              nearZ, farZ);
pMovie->SetProjectionMatrix3D(ortho);

```

5.1 AS3

ActionScript 3

PerspectiveProjection

Example:

```

import flash.display.Sprite;

var par:Sprite = new Sprite();
par.graphics.beginFill(0xFFCC00);
par.graphics.drawRect(0, 0, 100, 100);
par.x = 100;
par.y = 100;
par.rotationX = 20;

addChild(par);

var chRed:Sprite = new Sprite();
chRed.graphics.beginFill(0xFF0000);
chRed.graphics.drawRect(0, 0, 100, 100);
chRed.x = 50;
chRed.y = 50;
chRed.z = 0;

par.addChild(chRed);

var pp3:PerspectiveProjection=new PerspectiveProjection();
pp3.fieldOfView=120;
par.transform.perspectiveProjection = pp3;

```

6 Stereoscopic 3D

Scaleform 3Di UI 3D

Scaleform 3Di 3D PC NVIDIA 3D Vision Kit NVIDIA
Display
(Parallax Shift) Scaleform PS3
TinyPlayerStereo Scaleform
Apps \Samples \GfxPlayerTiny \GfxPlayerTinyPS3GcmStereo.cpp

6.1 NVIDIA 3D Vision

6.1.1 3D Vision

NVIDIA 3D Vision Kit 3Di UI, NVIDIA
3D 3D (Stereoscopic 3D
Test)

6.1.2

接下來以全屏模式啟動應用程式或 Scaleform Player。自動化 3D Vision 支援僅適用於全屏應用程式以及需要運用到 Direct3D Scaleform Player,因此,應用程式必須以全屏模式啟動。在 Scaleform Player 運用上,用戶需要應用到 command line argument -f。根據 NVIDIA 控制台的立體部分的設置,應用程式可以在禁用立體 3D 的情況下啟動。如果 3D 效果沒有立即出現,請按 (CTRL + T) 來切換到立體模式。

6.1.3

3D Vision
(Convergence)

3D Vision
(CTRL + F3) /(CTRL + F4)

IR

Scaleform Player
(CTRL + F5) (CTRL + F6)

Keyboard Shortcuts”
settings)

NVIDIA	“Stereoscopic 3D” “	“Set up stereoscopic 3D” ” (Enable the advanced in-game	“Set (CTRL + F7)
	(CTRL + F5)	(CTRL + F6)	

(CTRL + F5)

10 15 /

(CTRL + F5) (CTRL + F6)

0 ---

(CTRL + F7)

NVIDIA API 3D

3D

API
<http://developer.nvidia.com/object/nvapi.html>.

http://developer.download.nvidia.com/presentations/2009/SIGGRAPH/3DVision_Develop_Design_Play_in_3D_Stereo.pdf.

6.2 Scaleform Stereo API

	Scaleform 4.1	Render::HAL	API	3D
3D		HDMI 1.4		
	Scaleform			6.2.1

Scaleform 4 具有将视窗设置为并排或上下渲染单独的立体图像的能力。这为需要垂直或水平并列显示立体图像的设备提供了便利。

此功能可通过调用带有适当标志的 **Render::Viewport::SetStereoViewport** 来实现。

下面介绍添加到 SF4 的新的“视窗立体” (Viewport Stereo) 选项标志：

```
// 只用于显示硬件中的立体；使用相同大小缓冲区，但每只眼睛一半。
```

```
View_Stereo_SplitV      = 0x40,
View_Stereo_SplitH      = 0x80,
View_Stereo_AnySplit    = 0xc0,
```

```
void SetStereoViewport(unsigned display);
```

6.2.1 Scaleform Stereo

```
GRenderer::StereoParams s3DInfo;
s3DInfo.DisplayDiagInches = 46;      // 46 inch TV
s3DInfo.DisplayAspectRatio = 16.f / 9.f;
s3DInfo.Distortion = .75;           // 0 to 1
s3DInfo.EyeSeparationCm = 6.4;      // this will default 6.4cm
pRenderHAL->SetStereoParams(s3DInfo);
```

	Display	Display
Scaleform		

```
pMovie->Advance(delta);

pRenderer->GetHAL()->SetStereoDisplay(Render::StereoLeft);
pMovie->Display();

pRenderer->GetHAL()->SetStereoDisplay(Render::StereoRight);
pMovie->Display();
```

7

3Di

Scaleform SDK

3D Flash

SWF

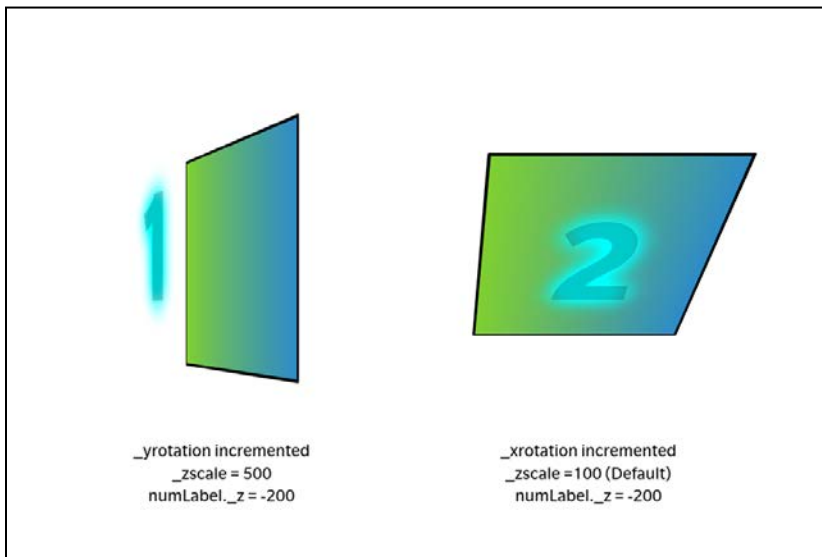
FLA

C:\Program Files (x86)\Scaleform\GFx SDK 4.4\Bin\Data\AS2 or

AS3\Samples



4 3DGenerator



5 3D



6 3D